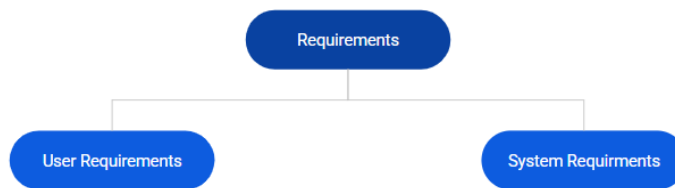


Requirements Classification



USER REQUIREMENTS

- What the users need
- Organized into user stories
- May be part of a use case



SYSTEM REQUIREMENTS

- What the system must do
- Include functional and non-functional requirements
- Describes software features and constraints

Mental health care patient information system (Mentcare)

User requirements definition

- 1.** The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

- 1.1** On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
- 1.2** The system shall generate the report for printing after 17.30 on the last working day of the month.
- 1.3** A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
- 1.4** If drugs are available in different dose units (e.g. 10mg, 20mg, etc.) separate reports shall be created for each dose unit.
- 1.5** Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

System stakeholders for the Mentcare system

- Mental health care patient information system (Mentcare)
 1. Patients whose information is recorded in the system and relatives of these patients.
 2. Doctors who are responsible for assessing and treating patients.
 3. Nurses who coordinate the consultations with doctors and administer some treatments.
 4. Medical receptionists who manage patients' appointments.
 5. IT staff who are responsible for installing and maintaining the system.
 6. A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
 7. Health care managers who obtain management information from the system.
 8. Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.
- **(1) A stockholding system for oil companies which keeps track of the amount of petrol (gas) at each of its sales outlets and automatically reorders new stock when the tanks fall below a certain level.**
- **Response:** Possible stakeholders are fuel delivery drivers, service station owners, service station sales staff, fuel supply operators, oil company management, oil company accountants, oil company buyers
- **(2) A train protection system which will automatically bring a train to a halt if it exceeds the speed limit for a track segment or if it goes through a red signal.**
- **Response:** Possible stakeholders train passengers, train driver, signallers, train operating company management, railway safety authorities, training staff

Levels of Requirements

- Business Requirements
- User Requirements
- Functional Requirements
- Non-Functional Requirements

Business Requirements

For example, a web application is designed and developed to offer better parcel delivery services to customers.

What is the business requirement ?

- Offer a smarter solution to measure, select, and ship parcels
- Provide capabilities to track and manage their delivery and pick-up services
- On-time delivery and customer feedback

User Requirements

- User requirements add further detail to the requirements
- User requirements are statements of **what services the system is expected to provide** to system users and the constraints under which it must operate.

Functional Requirements

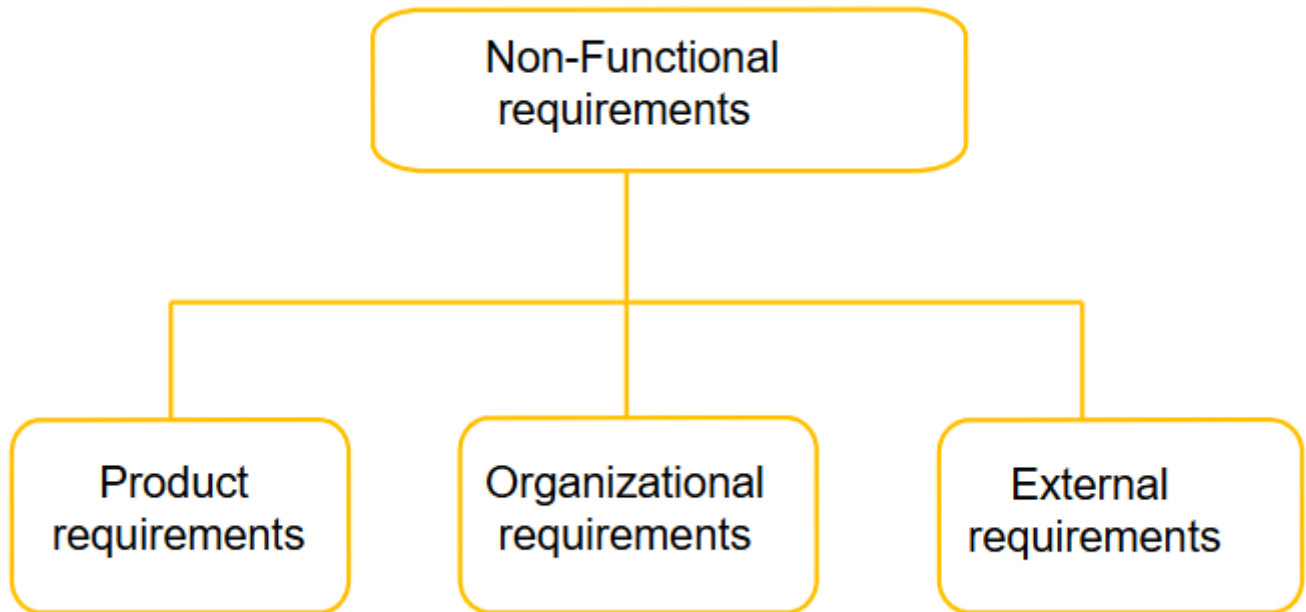
- These are statements of **services** the system should provide, **how the system should react to particular inputs**, and **how the system should behave in particular situations**
 - Depend on the type of software being developed
 - When expressed as user requirements, functional requirements should be written in natural language so that system users and managers can understand them.
 - Functional system requirements expand the user requirements and are written for system developers. They should describe the system functions, their inputs and outputs, and exceptions in detail.
- **In some cases, the functional requirements may also explicitly state what the system should not do**

Mentcare (Example)

1. A user shall be able to search the appointments lists for all clinics.
2. The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
3. Each staff member using the system shall be uniquely identified by his or her eight-digit employee number.

Non-Functional Requirements

- These are **constraints** on the services or functions offered by the system
- They include **timing constraints**, constraints on the **development process**, and constraints imposed by **standards**
- Non-functional requirements often apply to the system as a whole, rather than individual system features or services
- For example, to ensure that **performance requirements** are met in an **embedded system**, you may have to organize the system to **minimize communications** between components.



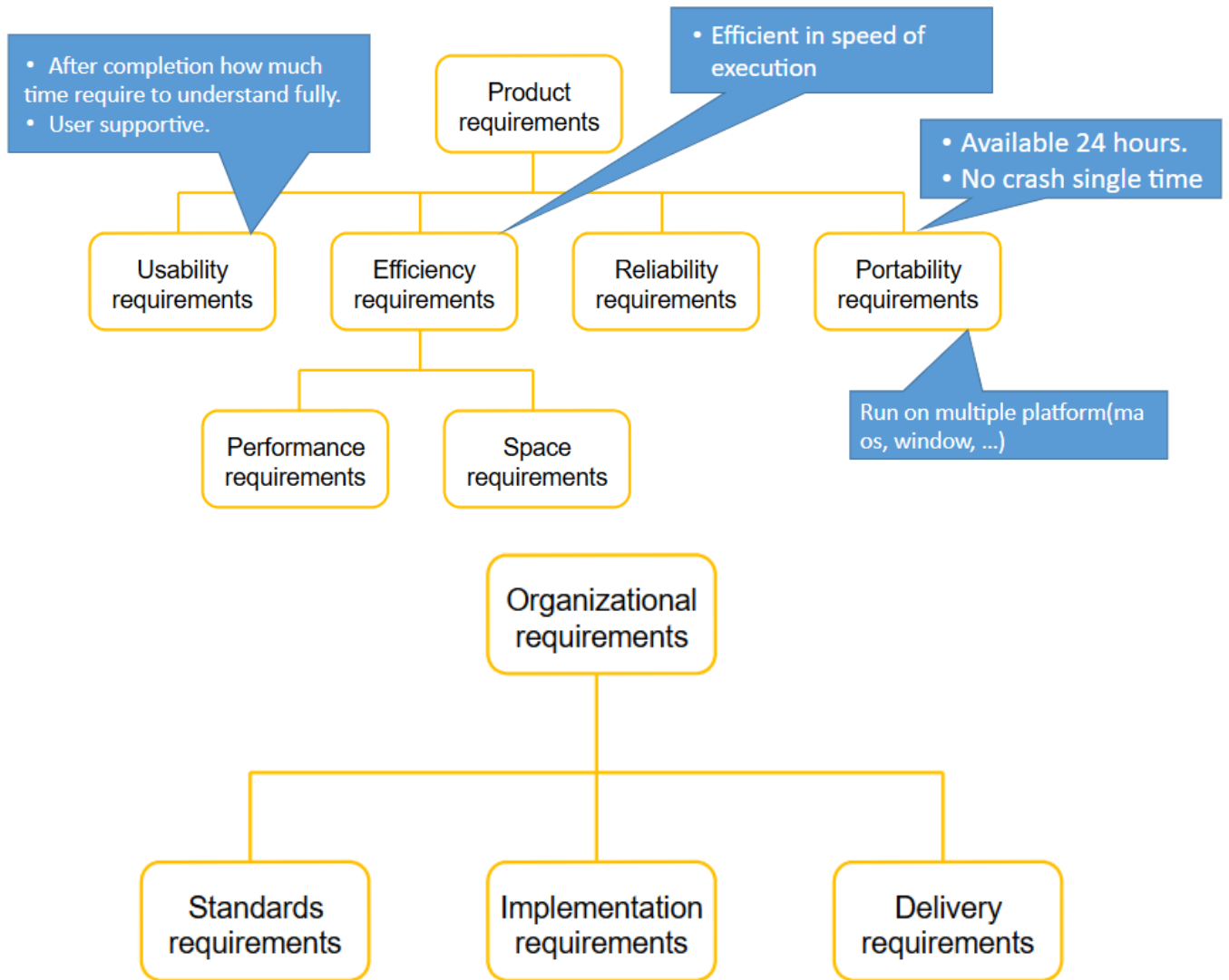
Product Requirements

- The system **shall** allow one hundred thousand hits per minute on the website

Reliability and performance

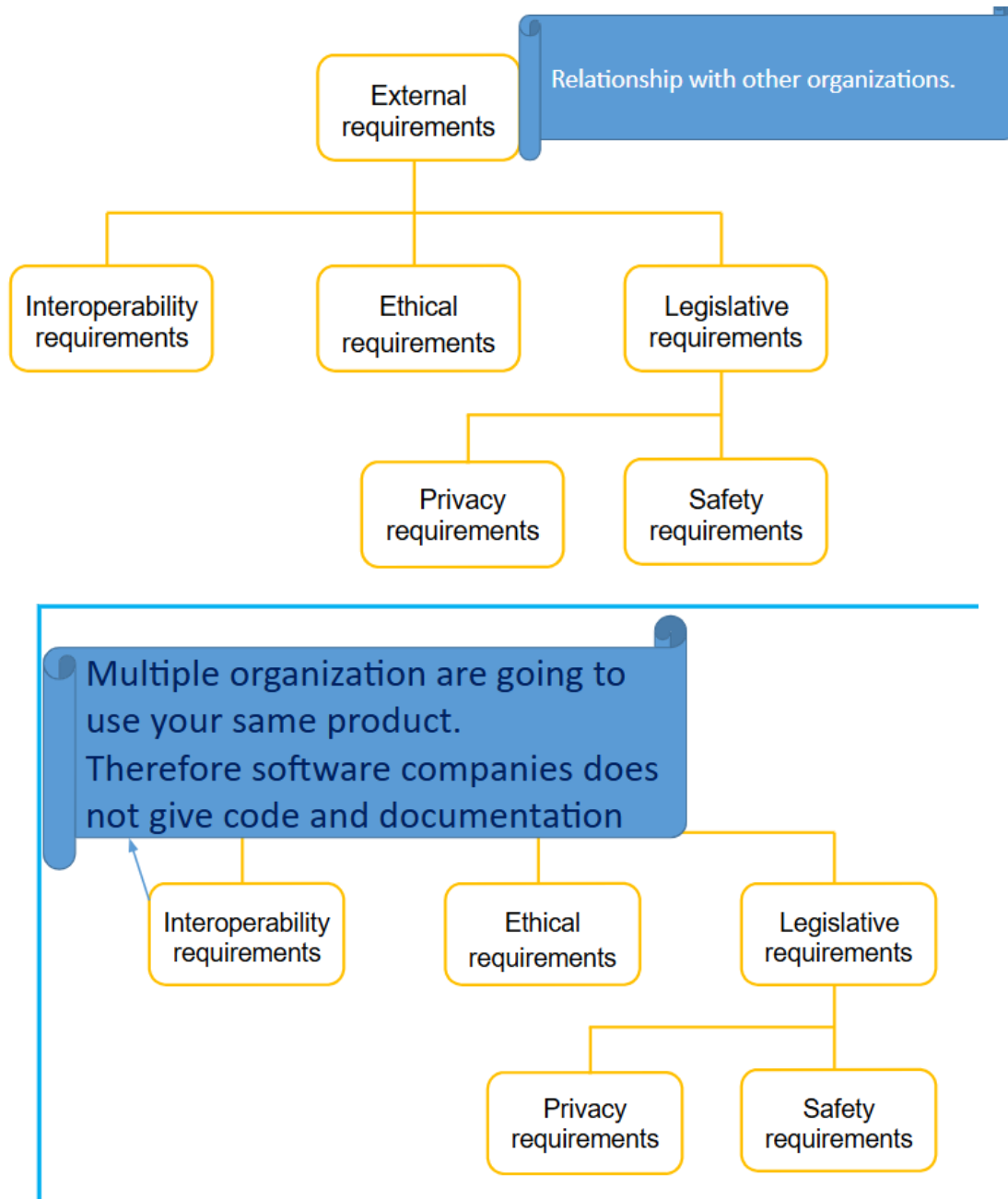
- The system **shall** not have down time of more than one second for continuous execution of one thousand hours

Mean time to failure



External Requirements

- The system **shall not** disclose any personal **information** about members of the library system to other members except system administrators
- The system **shall** comply with the local and national laws regarding the use of software tools



Non-Functional Requirements (NFRs)

1. Product Requirements

These define the inherent qualities and behaviours of the software product itself.

- **Usability Requirements:**
 - The system shall be user-supportive.
 - The system shall be easy to learn (After completion, a user shall be able to understand it fully within a short time).
- **Efficiency Requirements:**
 - The system shall be efficient in its speed of execution.
 - **Performance Requirements:** (Implied by efficiency) The system shall execute tasks and respond within a defined time frame.
 - **Space Requirements:** (To be defined) The system shall operate within specified hardware memory and storage constraints.

- **Reliability Requirements:**
 - The system shall be available 24 hours.
 - The system shall not crash a single time.
- **Portability Requirements:**
 - The system shall run on multiple platforms (e.g., macOS, Windows, Linux).

2. Organizational Requirements

These are constraints imposed by the developing organization's policies and processes.

- **Standards Requirements:** The product must adhere to internal architectural, coding, or documentation standards.
- **Implementation Requirements:** Mandates on specific technologies, programming languages, or databases to be used.
- **Delivery Requirements:** Constraints on the timing and method of delivery and installation.

3. External Requirements

These arise from factors external to the system and the developing organization.

- **Interoperability Requirements:** The system must work seamlessly with other specified systems or organizations.
- **Ethical Requirements:** The system must be designed and function in an ethically sound manner.
- **Legislative Requirements:** The system must comply with all relevant laws and regulations (e.g., data protection, financial).
 - **Privacy Requirements:** The system must protect user data and ensure confidentiality as defined by policy and law.
 - **Safety Requirements:** The system must not, under any condition, lead to a situation that could cause harm to users or the environment. That involves illicit content on the app.

Domain Requirements

- Requirements that come from the **application domain** and **reflect** fundamental **characteristics** of that application domain
- Can be **functional** or **non-functional**
- These requirements, sometimes, are not **explicitly mentioned**, as domain experts find it **difficult** to convey them. However, their absence can cause significant dissatisfaction

Domain Requirements

- Domain requirements can impose **strict constraints** on **solutions** 
- This is particularly true for **scientific** and **engineering** domains
- **Domain-specific** terminology can also cause **confusion**

Inverse Requirements

- They explain what the system **shall not do**.
- Many people find it convenient to describe their needs in this manner
- These requirements indicate the **indecisive** nature of customers about certain aspects of a new software product